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Minireview

Mitochondrial aminoacyl-tRNA synthetases in human disease

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ABSTRACT

Mitochondrial aminoacyl-tRNA synthetases (mtARSs) are essential in the process of transferring genetic information from mitochondrial DNA to the complexes of the oxidative phosphorylation system. These synthetases perform an integral step in the initiation of mitochondrial protein synthesis by charging tRNAs with their cognate amino acids. All mtARSs are encoded by nuclear genes, nine of which have recently been described as disease genes for mitochondrial disorders. Unexpectedly, the clinical presentations of these diseases are highly specific to the affected synthetase. Encephalopathy is the most common manifestation but again with gene-specific outcomes. Other clinical presentations include myopathy with anemia, cardiomyopathy, tubulopathy and hearing loss with female ovarian dysgenesis. Here we review the described mutation types and the associated patient phenotypes. The identified mutation spectrum suggests that only mutation types that allow some residual tRNA-charging activity can result in the described mtARS diseases but the molecular mechanisms behind the selective tissue involvement are not currently understood.

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1. Introduction

The accurate maintenance of the genetic code in the transmission of information from gene to the finished protein product is fundamental for all living cells. Protein synthesis is a highly complex process, which requires a large number of components to function effectively and accurately. Protein synthesis occurring in the mitochondria serves the mitochondrial oxidative phosphorylation system (OXPHOS), which is the major source of cellular ATP energy. Thirteen subunits of OXPHOS

Abbreviations: ARS, Aminoacyl-tRNA synthetase; CNS, Central nervous system; CSF, Cerebrospinal fluid; mtARS, Mitochondrial aminoacyl-tRNA synthetase; mtDNA, Mitochondrial DNA; MRI, Magnetic resonance imaging; OXPHOS, Oxidative phosphorylation.

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complexes are encoded by mitochondrial DNA (mtDNA) and therefore an intra-mitochondrial protein synthesis system is required to transfer the genetic information from mtDNA to the OXPHOS complexes [1]. Importantly, all protein components involved in mitochondrial translation are encoded by nuclear genes, including all translation factors, subunits of the mitoribosome and the mitochondrial aminoacyl-tRNA synthetases. These proteins cooperate with mtDNA-encoded ribosomal RNAs and transfer RNAs [2].

Aminoacyl-tRNA synthetases (ARSs) are a group of ancient enzymes that perform an integral step in the initiation of translation by charging tRNAs with their cognate amino acids [3]. These enzymes catalyze a two-step reaction where they first activate the cognate amino acid with ATP to form an aminoacyl-adenylate and then transfer the aminoacyl group of the aminoacyl-adenylate to the bound tRNA. Typically, aminoacyl–tRNA synthetases have a catalytic domain and an anticodon binding domain but some also have an editing domain for deacylating mischarged amino acids. The selectivity by which an ARS recognizes the specific amino acid and the correct tRNA is one of the important quality control mechanisms of protein synthesis [4].

The human ARSs for cytoplasmic and mitochondrial tRNA-charging are mainly encoded by distinct nuclear genes. (Nomenclature, *ARS* for the genes encoding cytoplasmic synthetases, *ARS2* for the genes encoding mitochondrial synthetases). The exceptions are *GARS* and *KARS*, which encode the glycyl- (GlyRS) and lysyl- (LysRS) tRNA synthetases in both compartments, respectively. Each amino acid is in principle recognized by its specific ARS, leading to the requirement of 20 ARSs in each cellular protein synthesis system. In human mitochondria, however, only 19 ARSs are present because Gln-tRNA^{Gln} is synthesized indirectly via misacylated Glu-tRNA^{Gln} through transamidation [5].

Recently, mutations in genes encoding mitochondrial ARSs (mtARSs) have emerged as a new cause of human disease, resulting in surprisingly tissue-specific phenotypes, although they are all expected to impair mitochondrial protein synthesis and thus affect the OXPHOS system. The described disease mutation types and the associated clinical symptoms are the subject of this review.

2. Tissue-specificity of human mtARS disorders

Genetic linkage studies, homozygosity mapping and whole-exome sequencing approaches have led to the identification of mtARS gene mutations in recessively inherited mitochondrial diseases (Table 1). Although mitochondrial OXPHOS dysfunction is known to lead to a multitude of disorders involving nearly any tissue of the body [6], the tissue- and cell type-specificity of mtARS diseases has been unexpected. MtDNA mutations, which are a common cause of OXPHOS disease, are often heteroplasmic because they occur in a multicopy genome. The varying heteroplasmy levels of mtDNA mutations in different tissues of a patient may partly explain the unpredictable phenotypic outcomes in those disorders but mutations causing mtARS diseases are present equally in all nuclear copies of the gene and should cause a constitutive failure in mitochondrial translation in all organs. Mutations in nine genes encoding mitochondria-specific ARSs have now been reported and among these disorders, encephalopathy is the most common phenotype while cardiomyopathy, tubulopathy, myopathy or sensorineural hearing loss with female ovarian dysgenesis is the main clinical presentation in others (Fig. 1). In addition, *GARS* and *KARS* mutations are found in patients with peripheral neuropathy but as these two genes encode for both the mitochondrial and cytoplasmic ARS, the evaluation of mitochondrial involvement in these phenotypes is more complicated.

From the fundamental function of mtARSs, one can elucidate that total loss-of-function is not compatible with extrauterine survival but through different mutational mechanisms, low enzyme activity may be maintained at levels that allow uterine development but result in specific diseases during postnatal life. A closer look at the exact molecular defects may explain some of the phenotypic variation in mtARS disorders.

3. Brain-specific mtARS diseases

Many OXPHOS diseases affect the brain because of the high energy demand of central nervous system (CNS) functions. Diseases caused by mtARS mutations, however, damage very specific neural cell types depending on the affected gene and result in specific clinical syndromes. Most of them have gene-specific brain magnetic resonance imaging (MRI) patterns.

3.1. DARS2

DARS2, encoding mitochondrial aspartyl-tRNA synthetase (mtAspRS), was the first identified mtARS disease gene [7]. Its mutations cause leukoencephalopathy with brain stem and spinal cord involvement and lactate elevation (LBSL), with varying age of onset from childhood to adulthood. The white matter changes in the patients are unique involving specific brain stem and spinal cord tracts. Elevated lactate suggesting a mitochondrial defect is observed only in the affected white matter and not in blood or cerebrospinal fluid (CSF). Other tissues or cultured cells of the LBSL patients do not show OXPHOS deficiency. The mutation pattern in typical LBSL patients is exceptional since nearly all patients are compound heterozygous for a splice site mutation in intron 2 and a missense mutation [7,8]. Recombinant mutant proteins containing the missense mutations were found severely deficient in charging tRNA^{Asp}, suggesting that those alleles are total loss-of-function [7]. By contrast, the intron 2 splice site mutations are thought to be “leaky” allowing some residual mtAspRS activity

Table 1
Clinical phenotypes of mutations in genes encoding mitochondrial aminoacyl-tRNA synthetases.

Gene	Mitochondrial protein	Main affected organ	Main clinical phenotype	Age at onset	Reference
AARS2	Alanyl-tRNA synthetase (mtAlaRS)	Heart	Hypertrophic cardiomyopathy	Infancy	[25]
DARS2	Aspartyl-tRNA synthetase (mtAspRS)	Brain	Leukoencephalopathy with brain stem and spinal cord involvement and lactate elevation (LBSL)	Childhood–adulthood	[7]
EARS2	Glutamyl-tRNA synthetase (mtGluRS)	Brain	Leukoencephalopathy with thalamus and brainstem involvement and high lactate (LTBL)	Infancy	[15]
FARS2	Phenylalanyl-tRNA synthetase (mtPheRS)	Brain	Alpers encephalopathy	Infancy	[19]
HARS2	Histidyl-tRNA synthetase (mtHisRS)	Cochlea Ovary	Progressive sensorineural hearing loss and ovarian dysgenesis (Perrault syndrome)	Childhood–adulthood	[28]
MARS2	Methionyl-tRNA synthetase (mtMetRS)	Brain	Autosomal recessive spastic ataxia with leukoencephalopathy (ARSAL)	Childhood–adulthood	[17]
RARS2	Arginyl-tRNA synthetase (mtArgRS)	Brain	Pontocerebellar hypoplasia type 6 (PCH6)	Infancy–childhood	[13]
SARS2	Seryl-tRNA synthetase (mtSerRS)	Kidney	Tubulopathy (hyperuricemia, metabolic alkalosis), pulmonary hypertension, and progressive renal failure (HUPRA)	Infancy	[23]
YARS2	Tyrosyl-tRNA synthetase (mtTyrRS)	Muscle	Myopathy, lactic acidosis, and sideroblastic anemia (MLASA)	Infancy–Childhood	[20]
GARS	Glycine-tRNA synthetase (mtGlyRS)	Nerve	Charcot-Marie-Tooth disease (CMT), dominant	Childhood–adulthood	[29]
KARS	Lysyl-tRNA synthetase (mtLysRS)	Nerve	Charcot-Marie-Tooth disease (CMT), recessive and dominant	Childhood–adulthood	[30]

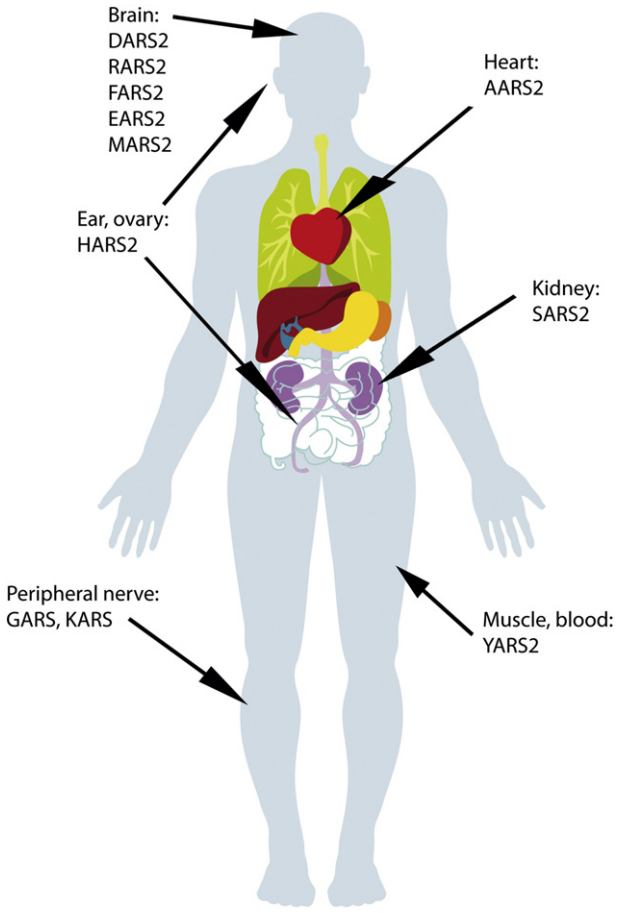


Fig. 1. Main affected tissues in disorders affecting mitochondrial aminoacyl-tRNA synthetases.

to maintain mitochondrial translation [9]. Apparently, this remaining activity is sufficient in most cell types but not in the specific white matter tracts. As an explanation for the tissue-specificity, differences in mRNA splicing machinery and splicing efficiency between neuronal and other cell types have been suggested [9]. Since all firstly identified patients had compound heterozygous mutations, it was suspected that homozygosity for *DARS2* missense mutations would be lethal and homozygosity for the splicing mutation would be too mild to manifest as a disease phenotype. Some reports of homozygous *DARS2* mutations have now been published and unexpectedly, the homozygous splicing mutation appears to result in more severe phenotypes [10,11] and a homozygous missense mutation in a milder phenotype [12]. Identification of additional homozygous mutations and their enzyme activity measurements would be informative.

3.2. RARS2

Mutations in *RARS2*, encoding mitochondrial arginyl-tRNA synthetase (mtArgRS), were first identified in patients suffering from a severe infantile pontocerebellar hypoplasia leading to early death [13]. Progressive atrophy was seen in cerebellum, pons, cerebral cortex and white matter. Patients described later had neonatal-onset epileptic encephalopathy with similar MRI findings [14]. While the symptoms were confined to CNS, skeletal muscle homogenates and even skin fibroblasts of some patients showed OXPHOS complex deficiencies and slightly elevated blood lactate was measured. The reported *RARS2* mutations were all compound heterozygous, with hypomorphic splicing alleles in some patients and missense mutations in others. The mutations have been shown to cause reduced cellular abundance of *RARS2* mRNA or mtArgRS protein, $tRNA^{Arg}$ stability or mtArgRS activity but no clear correlation to clinical severity has been ascribed to these alterations [14].

3.3. EARS2

Mutations in *EARS2*, encoding mitochondrial glutamyl-tRNA synthetase (mtGluRS), were described in LTBL (leukoencephalopathy with thalamus and brainstem involvement and high lactate) patients [15]. Again, the brain MRI features were unique to these patients, showing symmetrical abnormalities of cerebral white matter, thalamus, midbrain, pons, medulla oblongata and cerebellar white matter. Elevated blood lactate and OXPHOS defects in muscle and skin fibroblasts were observed. The disease had infantile onset and rapid progression with a striking partial recovery and regaining of milestones in some patients and no improvement in others. The authors suggested this to indicate that *EARS2* mutations particularly affect brain development with a chance of recovery during further brain maturation if the initial damage was not too severe and nerve cell survival was substantial. All patients with *EARS2* mutations were compound heterozygous typically with a severe mutation combined with a milder mutation, indicating that only a certain level of residual mtGluRS activity can lead to the LTBL phenotype. This conclusion is further supported by the report of one patient with a homozygous *EARS2* mutation, who suffered from a severe infantile multisystem disease involving the brain and liver [16].

3.4. MARS2

MARS2, encoding mitochondrial methionyl-tRNA synthetase (mtMetRS), was reported to underlie the ARSAL (autosomal recessive spastic ataxia with leucoencephalopathy) syndrome [17]. In addition to cerebellar ataxia and spasticity, brain MRI showed cerebellar atrophy, which in some cases was associated with cortical atrophy, leucoencephalopathy and corpus callosum thinning [18]. The age of onset varied greatly, between 2 and 59 years. Unlike the other mtARS diseases, ARSAL is caused by complex genomic rearrangements in *MARS2* including gene duplication events that result in truncated and/or reduced amount of mtMetRS in patient cells. If the rearrangement included a deletion–duplication event instead of duplication only, the age of disease onset was lower. Mitochondrial translation and OXPHOS Complex I defects were observed in the lymphoblastoid lines of the clinically most severely affected patients.

3.5. FARS2

Mutations in *FARS2*, encoding mitochondrial phenylalanyl-tRNA synthetase (mtPheRS), cause a fatal infantile epileptic encephalopathy with lactacidosis [19]. The brain MRI findings of these patients showed extensive cerebral atrophy and signal changes in putamina and caudate nuclei with relative sparing of the cerebellum. Neuropathological findings were consistent with Alpers disease, showing laminal cortical necrosis and severe cortical atrophy. Blood lactate was elevated and muscle showed OXPHOS deficiency but patient fibroblasts displayed no phenotype. The described patient mutations were either compound heterozygous missense mutations or a homozygous missense mutation. Characterization of the recombinant mutant mtPheRS proteins showed that each mutation affected the synthetase function in a different manner, by reducing ATP, phenylalanyl or tRNA binding. The mutations significantly reduced the tRNA charging capacity of mtPheRS but were not fully loss-of-function alleles.

4. mtARS diseases affecting mainly other tissues than the brain

The rest of the mtARS disorders are not CNS-specific but together their phenotypes involve the same range of affected tissues that are typically found in other OXPHOS diseases. Nevertheless, each mtARS disorder in this group differs from all others in their distinctive tissue-involvement

and clinical outcome. Few affected patients have so far been described for these disease entities, enabling less informative genotype-phenotypes correlations and leaving room for the involvement of these genes in a wider range of phenotypes. Although homozygous mutations that severely disrupt the synthetase function have been identified in some of these mtARS diseases, yet the main affected organ is gene-specific.

4.1. YARS2

YARS2, encoding the mitochondrial tyrosyl-tRNA synthetase (mtTyrRS), has been found mutated in patients with myopathy, lactic acidosis and sideroblastic anemia (MLASA) [20]. The muscle biopsy sample of the patients and patient myoblasts showed OXPHOS deficiency while patient fibroblasts did not. Two homozygous *YARS2* missense mutations affecting the catalytic domain of mtTyrRS have been reported, both affecting the stability of the protein and resulting in a generalized mitochondrial translation defect in patient myoblasts [20,21]. Functional assays with recombinant protein showed reduced catalytic activity for mutant mtTyrRS with reduced affinity for $tRNA^{Tyr}$. Why such a defect leads to a combined myopathy and anemia is not known but it should be noted that MLASA can also be caused by mutations in *PUS1*, encoding pseudouridylate synthase 1, which may be responsible for pseudouridylation of $tRNA^{Tyr}$ [22]. This may suggest that $tRNA^{Tyr}$ and not the general mitochondrial translation defect is specifically involved in the development of the combined phenotype.

4.2. SARS2

Homozygous missense mutation in *SARS2*, encoding mitochondrial seryl-tRNA synthetase (mtSerRS), was described in patients with tubulopathy (hyperuricemia, metabolic alkalosis), primary pulmonary hypertension, and progressive renal failure in infancy (HUPRA syndrome) [23]. High blood and CSF lactate were measured and muscle biopsy showed signs of OXPHOS deficiency. The patients died within the first 14 months of life. mtSerRS is exceptional among the mtARSs (in addition to mtLeuRS) because it charges two tRNA isoacceptors, $tRNA^{Ser}_{AGY}$ and $tRNA^{Ser}_{UCN}$ [24]. Patient lymphoblasts had significantly reduced levels of $tRNA^{Ser}_{AGY}$ but normal $tRNA^{Ser}_{UCN}$ levels, showing that this specific *SARS2* mutation only affected the charging of one tRNA isoacceptor.

4.3. AARS2

AARS2, encoding the mitochondrial alanyl-tRNA synthetase (mtAlaRS), was found to be a disease gene for early-onset fatal hypertrophic cardiomyopathy with lactacidosis [25]. The described patients died perinatally or within the first 10 months of life. Cardiomyopathy was the prominent clinical finding but in autopsy OXPHOS deficiency was also observed in brain and muscle in addition to heart. Patient fibroblasts or myoblasts did not, however, display an OXPHOS defect. The identified *AARS2* mutations were either homozygous or heterozygous but so far all described patients have had a mutation leading to p.R592W alteration in mtAlaRS [25,26]. A patient with the homozygous mutation lived 10 months whereas patients with a combined p.R592W and a likely total loss-of-function defect in the catalytic domain died perinatally or in utero [25]. This synthetase is unique among the human mtARSs because it contains an editing domain in addition to the catalytic domain. AlaRSs through all kingdoms of life have a problem in the distinction of alanine from serine and glycine because all three amino acids fit the binding pocket equally [27]. Therefore, the editing domain functions in deacylating the mischarged tRNAs, which otherwise could be detrimental to mitochondrial translation. The p.R592W mutation is located at the surface of the editing domain, with predicted effect on the binding/recognition of the mischarged tRNAs [25]. Thus the pathogenic mechanism of *AARS2* mutations may be mitochondrial mistranslation, which would be different from the other mtARS defects, but the hypothesis has not been experimentally verified. Why cardiomyopathy would be specifically caused by mistranslation while translation deficiency caused by the other mtARS defects seems to have little effect on heart, a tissue with high energy demand, is an open question.

4.4. HARS2

Mutations in *HARS2*, encoding the mitochondrial histidyl-tRNA synthetase (mtHisRS), have been described in one family with five members affected with Perrault syndrome [28]. The disease is characterized by sensorineural hearing loss and female ovarian dysgenesis. Patients were found to be compound heterozygous for a missense mutation and a mutation affecting mRNA splicing. The missense mutation was predicted to critically affect tRNA binding and was shown to severely reduce the catalytic activity in an assay using recombinant proteins. Thus, in addition to *DARS2* and *RARS2* mutations, the tissue-specificity of the mtHisRS defect may be influenced by splicing efficiency in different cell types. The identification of other patients with *HARS2* mutations would confirm the association to Perrault syndrome.

5. *GARS* and *KARS* mutations in peripheral neuropathies

Charcot-Marie-Tooth (CMT) disease is a heterogeneous group of peripheral neuropathies characterized by loss of muscle tissue and touch sensation predominantly in feet and legs. Mutations in *GARS* [29] and *KARS* [30] are largely believed to result in peripheral neuropathy phenotypes, similar to those caused by *AARS*, *YARS* and *HARS* mutations [31–33], because they are all involved in the cytoplasmic translation. Defective mitochondrial dynamics is known to be the most frequent cause of axonal CMT, triggered by mutations in MFN2 encoding a protein essential for mitochondrial fusion. The disturbances that *GARS* or *KARS* mutations may cause for mitochondrial function have not, however, been fully investigated. In contrast to the recessive mtARS diseases, *GARS* mutations (as well as *AARS*, *YARS* and *HARS*) are dominantly inherited and experimental evidence shows toxicity in peripheral neurons through dominant negative effect of the mutant GlyRS protein [34]. On the other hand, recessively inherited compound heterozygous *KARS* mutations have been described in an unusual combination of intermediate CMT, developmental delay and dysmorphic features [30]. Furthermore, in vitro LysRS aminoacylation activity of the mutant proteins was minimal, suggesting that a mitochondrial translation defect may also be present. Finally it should be noted that axonal neuropathy has also been reported in some patients with *DARS2* mutations [8].

6. Conclusions/perspectives

The basic conclusion of the genotype–phenotype correlations in mtARS diseases is that only certain types of mutations that allow some residual tRNA-charging activity can cause these diseases. The function of the synthetases is essential for development thus preventing the identification of patients with full loss-of-function alleles. On the other hand, too mild combinations of mutations are not found either, because surprisingly low synthetase activity is able to maintain OXPHOS function in all tissues. All identified mutations or their combinations retain some enzyme activity but the reason for why the elsewhere sufficient residual activity of a certain synthetase is not enough in a specific cell type is presently completely unclear. The *EARS2* mutations interestingly suggest that the question may even be of a specific time in development, which can later be overcome. Tissue-specific differences in mitochondrial chaperone activities may play a role in determining the stability of some mutants. In regard to the tissue-specificity, the altered levels of particular uncharged tRNAs and amino acids and their potential as signaling molecules are also interesting aspects.

Functional studies of many of the mtARS defects are hindered by the lack of manifestation in cultured cells such as fibroblasts. Induced

pluripotent stem cells derived from patient cell lines, differentiated into various cell types, may offer advantages in future studies. Mouse models may also be useful in the investigation of these diseases but to date none have been reported (with the exception of *GARS* [35]). Challenges inevitably arise in generating mouse models because most of these disorders are caused by compound heterozygous mutations or even potentially "leaky" splice site mutations that are not the most straight-forward to model in mice. For example, a full knock-out mouse would not mimic the human disease in which residual mtARS activity is present. A fly model for ARSAL has been described with partial loss-of-function mutations affecting the *Drosophila* mtMetRS [17]. The flies that escaped lethality had reduced lifespan and showed degeneration in muscle and brain, partial OXPHOS Complex I deficiency, signs of increased reactive oxygen species and upregulation of the mitochondrial unfolded protein response. Antioxidants fed to the flies suppressed some of the observed phenotypes. Follow-up studies on the fly model will hopefully prove beneficial for understanding ARSAL.

We may be currently having a too narrow view on the tissue-specificity issue. Research on ARSs has recently revealed remarkable noncanonical functions for the cytoplasmic synthetases [36]. One example is SerRS, which is essential for vascular development in vertebrates through its alternative localization to the nucleus and function in gene expression regulation of vascular endothelial growth factor A [37]. MtARSs could also have additional functions that are indispensable in specific cell types or developmental stages. Nine cytoplasmic ARSs with auxiliary proteins form a multi-synthetase complex [38] and for the mtARSs mtTyrRS has already been demonstrated to be part of a high-molecular weight complex [21]. Whether these complexes play a role in some aspects of mtARS disease remains to be studied.

In general, the mitochondrial ARSs are relatively poorly characterized and future studies will provide important knowledge of the functions of these synthetases. Examination of the identified *ARS2* mutations has revealed some clues for the understanding of the molecular mechanisms of these rare diseases. One unifying explanation for the gene-specific clinical presentations is not likely to be found but generation of appropriate disease models and additional detailed studies of the patients are a good start for increasing our understanding of these diseases.

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